INFORMATION SECURITY

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Project Proposal:
Bad USB / HID Keystroke Injection & Detection

Introduction

USB devices can act as Human Interface Devices (HID) like keyboards or mice. Malicious devices can exploit this to inject keystrokes automatically, execute commands, or compromise system security.

This project demonstrates a system that monitors keystroke activity and detects unusual patterns that indicate potential HID-based attacks. It provides real-time alerts and notifications for suspicious activity, helping users understand and respond to threats from unauthorized USB devices.

Problem Statement / Project Idea

Modern computers are vulnerable to Bad USB and automated keystroke injection attacks. Current security systems often fail to detect these subtle threats because the devices appear legitimate.

Objectives:

- Monitor all keystrokes coming from USB-connected devices.
- Detect abnormal typing patterns, including unusually fast or perfectly rhythmic keystrokes.
- Alert users immediately when suspicious activity is detected.
- Provide a safe and controlled simulation to understand HID-based attack threats.

Scope

The system will:

- Continuously monitor HID input devices for anomalous activity.
- Identify abnormal typing patterns (speed and variance anomalies).
- Generate visual and popup alerts for detected threats.

- Maintain a log of suspicious activity with timestamps for review.

Limitations:

- Focuses only on HID-based attacks, not other USB malware.
- Detects abnormal input patterns, but cannot prevent physical device insertion.

Significance / Expected Outcome

This project:

- Highlights the security risks posed by malicious USB devices.
- Helps users and organizations detect potential HID-based intrusions in real time.
- Acts as a practical demonstration for education and awareness on modern USB threats.

Conclusion

HID keystroke injection attacks pose a hidden threat to system security. It detects suspicious input patterns in real time, alerting users to potential risks and helping mitigate unauthorized USB-based attacks.